

AI-Powered Resume Ranker

Project	AI-Powered Resume Ranker
Domain	Machine Learning & NLP
Tools Used	Python, SpaCy, Scikit-learn, Flask, PyPDF2

Introduction

The AI-Powered Resume Ranker is a smart application that automates the process of resume screening using Machine Learning and Natural Language Processing techniques. It helps recruiters shortlist candidates efficiently by comparing resumes with job descriptions.

Abstract

This project extracts text from resumes, preprocesses it using SpaCy, and transforms it into numerical vectors using TF-IDF. Cosine similarity is then applied to measure the relevance of resumes to job descriptions, enabling automatic ranking.

Tools & Technologies

Python is used as the programming language. SpaCy is used for NLP preprocessing. Scikit-learn is used for TF-IDF and cosine similarity. Flask is used to build the web interface, and PyPDF2 is used for extracting text from PDF resumes.

Methodology

1. Extract text from PDF resumes. 2. Preprocess text using SpaCy (remove stopwords, punctuation, lemmatization). 3. Convert text into numerical form using TF-IDF. 4. Compute similarity using cosine similarity. 5. Rank resumes based on similarity scores. 6. Display results via Flask UI and allow download.

Results & Discussion

The system successfully ranks resumes based on their relevance to the job description. Higher scores indicate better matching resumes. The results are displayed in a structured table and can be downloaded as a CSV file.

Advantages

- Reduces manual effort in resume screening. - Improves efficiency and speed. - Provides objective ranking based on data. - Easy to use web interface.

Future Scope

Future improvements include integrating advanced NLP models like BERT, adding skill extraction, enhancing UI/UX, and connecting the system with job portals.

Conclusion

This project demonstrates how Machine Learning and NLP can be applied to real-world problems like recruitment. It improves hiring efficiency and provides a scalable solution for candidate evaluation.